

# **COMSATS UNIVERSITY ISLAMABAD**

## **ATTOCK CAMPUS**



## **DEPARTMENT OF COMPUTER SCIENCE**

|                   |              |
|-------------------|--------------|
| <b>COURSE</b>     | SQE Lab      |
| <b>INSTRUCTOR</b> | Ms. Munazza  |
| <b>TASK</b>       | Assignment 4 |

### **Submitted By:**

|               |                         |
|---------------|-------------------------|
| <b>NAME</b>   | <b>REGISTRATION NO.</b> |
| JAVERIA IRUM  | FA23-BSE-063            |
| MARYAM FATIMA | FA23-BSE-007            |
| AYESHA        | FA23-BSE-032            |

# Academic Task Tracker System

## Introduction:

Academic Task Tracker System is a Python-based software application developed to help students manage their academic tasks. This system allows users to manage assignments, quizzes, mids, terminals, and project activities. Users can add tasks, view tasks, search tasks, update task status, and delete tasks. The system helps students stay organized and track their academic progress

## Tools Used:

Python – Used for system development.

Visual Studio Code – Used for writing and running code.

SonarQube – Used for code quality analysis.

Snyk – Used for security analysis.

[GitHub](#) – Used for version control and project hosting.

## Core Features:

1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit System

## Manual Test Cases:

| Test Case ID | Feature     | Input                   | Expected Output         | Actual Output           | Status |
|--------------|-------------|-------------------------|-------------------------|-------------------------|--------|
| TC-01        | Add Task    | PF, Assignment, Pending | Task added successfully | Task added successfully | Pass   |
| TC-02        | View Tasks  | Press 2                 | Tasks displayed         | Tasks displayed         | Pass   |
| TC-03        | Search Task | PF                      | Task Found              | Task Found              | Pass   |

|       |               |               |                |                |      |
|-------|---------------|---------------|----------------|----------------|------|
| TC-04 | Update Status | PF, Completed | Status updated | Status updated | Pass |
| TC-05 | Delete Task   | PF            | Task deleted   | Task deleted   | Pass |
| TC-06 | Exit          | Press 6       | Program closed | Program closed | Pass |

## Bug Report:

| Bug ID | Feature     | Problem Found                             | Expected Behavior             | Actual Behavior          | Status |
|--------|-------------|-------------------------------------------|-------------------------------|--------------------------|--------|
| BUG-01 | View Tasks  | After restarting program, tasks disappear | Tasks should remain available | Tasks list becomes empty | Fixed  |
| BUG-02 | Search Task | Wrong course name entered                 | Show not found message        | Task not found shown     | Fixed  |

## Program execution screenshots:

```

PS C:\Users\JHP\OneDrive\Desktop\Academic_Task_Tracker> git push origin main --force
PS C:\Users\JHP\OneDrive\Desktop\Academic_Task_Tracker> python main.py

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 1
Enter course name: Sqa Lab
Enter task type (Assignment/Quiz/Mid/Terminal/Project): Project
Enter status (Pending/Ongoing/Completed): Completed
Task added successfully!

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 2

===== All Tasks =====
Course: Sqa Lab
Task Type: Project
Status: Completed
=====

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status

```

```
File Edit Selection View Go Run ... Academic_Task_Tracker Update
EXPLORER PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ACADEMIC_TASK_TRACKER
  > .sonar
  main.py
  requirements.txt
PS C:\Users\HP\OneDrive\Desktop\Academic_Task_Tracker> python main.py

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 3
Enter course name to search: Sqe Lab

Task Found!
Course: Sqe Lab
Task Type: Project
Status: Completed

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 4
Enter course name to update: SQE LAB
Enter new status: PROJECT IS COMPLETED
Status updated successfully!

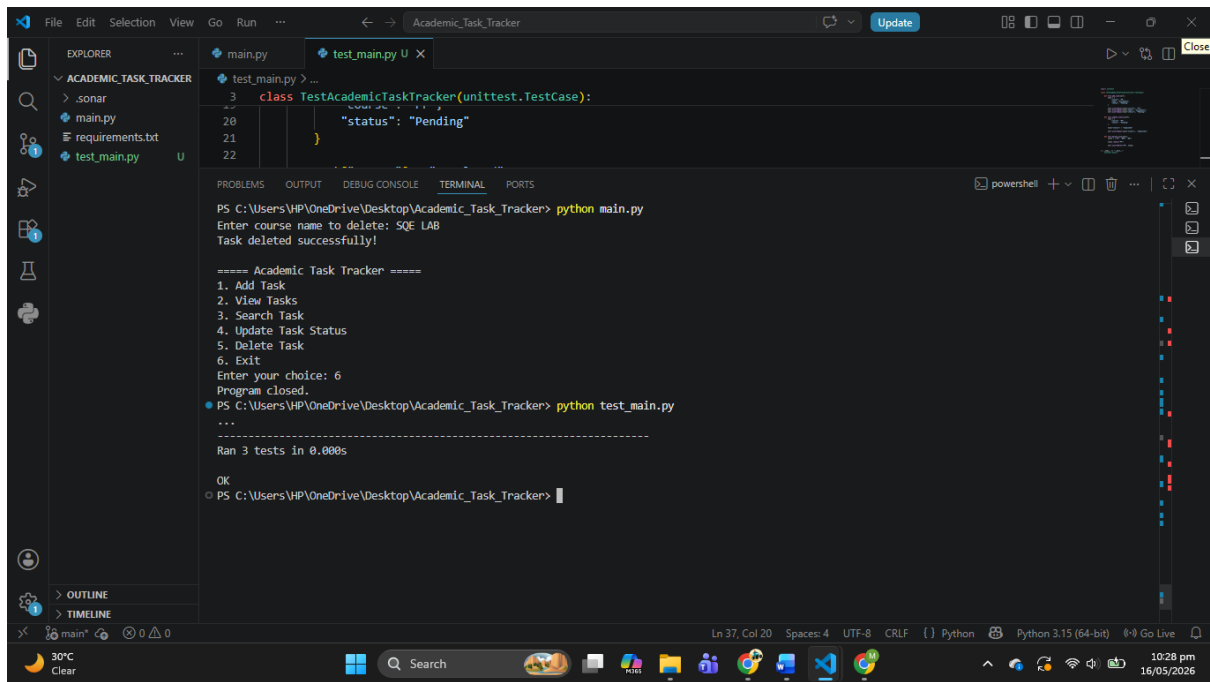
===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
```

```
File Edit Selection View Go Run ... Academic_Task_Tracker Update
EXPLORER PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ACADEMIC_TASK_TRACKER
  > .sonar
  main.py
  requirements.txt
PS C:\Users\HP\OneDrive\Desktop\Academic_Task_Tracker> python main.py
Status: Completed

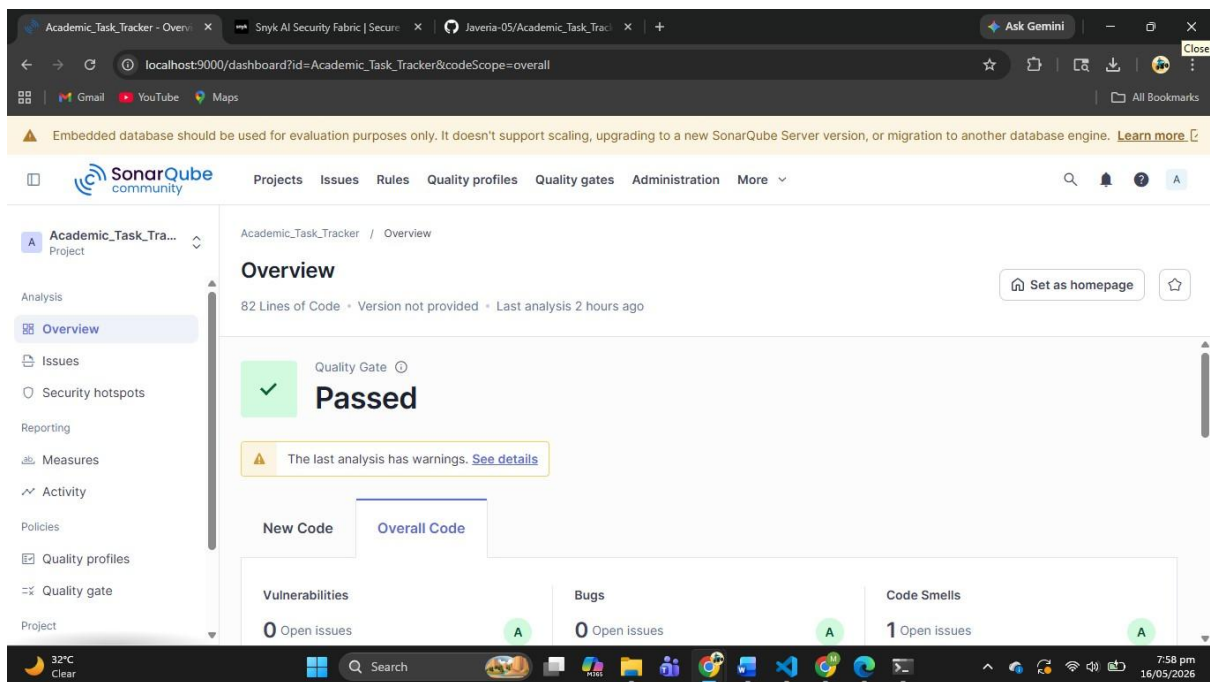
===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 4
Enter course name to update: SQE LAB
Enter new status: PROJECT IS COMPLETED
Status updated successfully!

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 5
Enter course name to delete: SQE LAB
Task deleted successfully!

===== Academic Task Tracker =====
1. Add Task
2. View Tasks
3. Search Task
4. Update Task Status
5. Delete Task
6. Exit
Enter your choice: 6
Program closed.
PS C:\Users\HP\OneDrive\Desktop\Academic_Task_Tracker>
```



## SonarQube analysis screenshots:



Academic\_Task\_Tracker - Overview | Snyk AI Security Fabric | Secure | Javeria-05/Academic\_Task\_Tracker | Ask Gemini

localhost:9000/dashboard?id=Academic\_Task\_Tracker&codeScope=overall

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)

SonarQube community

Projects Issues Rules Quality profiles Quality gates Administration More

Academic\_Task\_Tracker / Overview

Overview

82 Lines of Code • Version not provided • Last analysis 2 hours ago

Set as homepage

|                                                                 |                                                  |                                              |
|-----------------------------------------------------------------|--------------------------------------------------|----------------------------------------------|
| <b>Vulnerabilities</b><br>0 Open issues<br>A                    | <b>Bugs</b><br>0 Open issues<br>A                | <b>Code Smells</b><br>1 Open issues<br>A     |
| <b>Accepted issues</b><br>0<br>Valid issues that were not fixed | <b>Coverage</b><br>0.0%<br>On 76 lines to cover. | <b>Duplications</b><br>0.0%<br>On 124 lines. |
| <b>Security Hotspots</b><br>0<br>A                              |                                                  |                                              |

32°C Clear 8:09 pm 16/05/2026

Academic\_Task\_Tracker - Issues | Snyk AI Security Fabric | Secure | Javeria-05/Academic\_Task\_Tracker | Ask Gemini

localhost:9000/project/issues?issueStatuses=OPEN%2CCONFIRMED&types=CODE\_SMELL&tid=Academic\_Task\_Tracker

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)

SonarQube community

Projects Issues Rules Quality profiles Quality gates Administration More

Academic\_Task\_Tracker / Issues

Issues

My Issues All

Bulk Change Select issues Navigate to issue 1 issues 6min effort

main.py

Define a constant instead of duplicating this literal "Task not found." 3 times.

Code Smell Critical design

Open Not assigned L57 • 6min effort • 2 hours ago

1 of 1 shown

Community Build

32°C Clear 8:09 pm 16/05/2026

# SonarQube Analysis Findings

Code was analyzed using SonarQube.

## Issue 1: Code Smell

Repeated strings were found in the source code.

### Suggested Fix:

Use constants or variables instead of writing the same text multiple times.

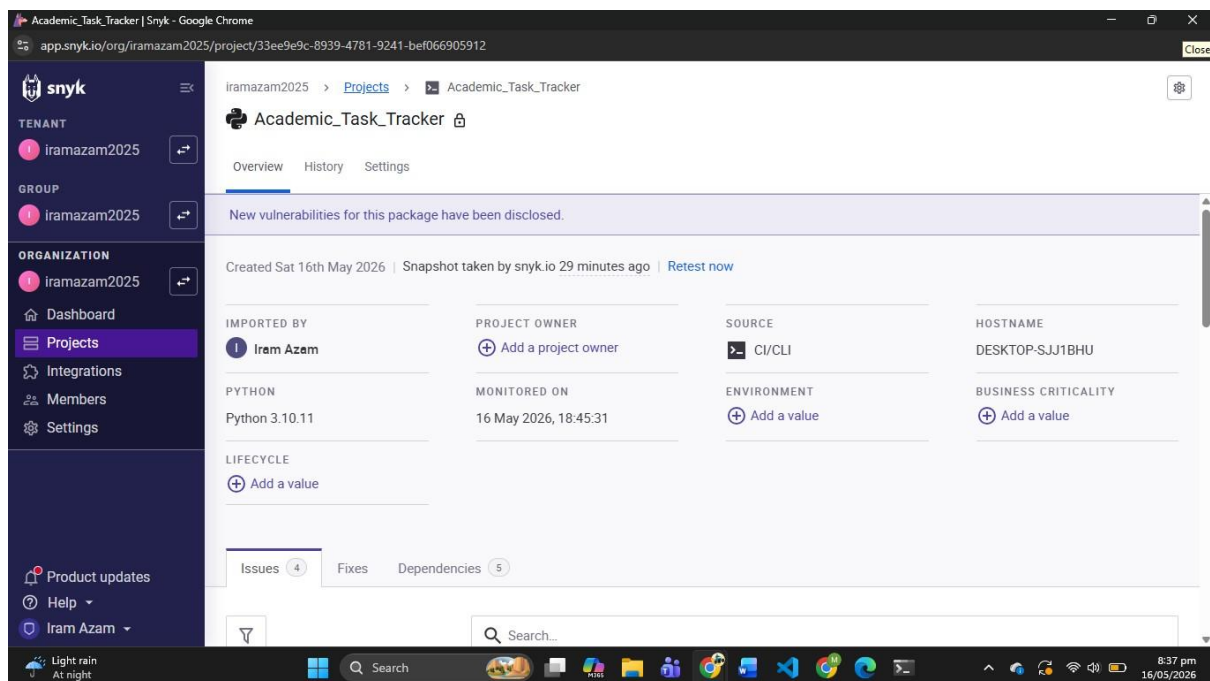
## Issue 2: Maintainability Issue

Some code blocks were repetitive.

### Suggested Fix:

Reusable functions should be used to improve code quality.

## Snyk security scan screenshots:





Academic\_Task\_Tracker | Snyk - Google Chrome  
app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

**snyk**  
TENANT  
iramazam2025  
GROUP  
iramazam2025  
ORGANIZATION  
iramazam2025  
Dashboard  
Projects  
Integrations  
Members  
Settings  
Product updates  
Help  
Iram Azam

iramazam2025 > Projects > Academic\_Task\_Tracker

Academic\_Task\_Tracker

Overview History Settings

Issues (4) Fixes Dependencies (5)

Search...

4 of 4 issues Group by none Sort by highest priority score

**H** **urllib3 - Decompression Bomb** SCORE **731**  
VULNERABILITY | ...

Introduced through requests@2.32.3 Exploit maturity NO KNOWN EXPLOIT  
Fixed in urllib3@2.7.0

Show more detail

Ignore

SEVERITY  
☐ Critical 0  
☐ High 2  
☐ Medium 2  
☐ Low 0

PRIORITY SCORE  
Scored between 0 - 1000

\*FIXED IN\* AVAILABLE  
☐ Yes 4

30°C Clear 10:46 pm 16/05/2026

Academic\_Task\_Tracker | Snyk - Google Chrome  
app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

**snyk**  
TENANT  
iramazam2025  
GROUP  
iramazam2025  
ORGANIZATION  
iramazam2025  
Dashboard  
Projects  
Integrations  
Members  
Settings  
Product updates  
Help  
Iram Azam

iramazam2025 > Projects > Academic\_Task\_Tracker

Academic\_Task\_Tracker

Overview History Settings

☐ No 0

COMPUTED FIXABILITY  
☐ Fixable 4  
☐ Partially fixable 0  
☐ No supported fix 0

EXPLOIT MATURITY  
☐ Mature 0  
☐ Proof of concept 1  
☐ No known exploit 3  
☐ No data 0

CVSS 4 EXPLOIT MATURITY  
☐ Attacked 0

**H** **urllib3 - Insertion of Sensitive Information Into Sent Data** SCORE **696**  
VULNERABILITY | ...

Introduced through requests@2.32.3 Exploit maturity NO KNOWN EXPLOIT  
Fixed in urllib3@2.7.0

Show more detail

Ignore

**M** **requests - Insertion of Sensitive Information Into Sent Data** SCORE **606**  
VULNERABILITY | ...

Introduced through requests@2.32.3 Exploit maturity PROOF OF CONCEPT  
Fixed in requests@2.32.4

Show more detail

Rain coming 9:12 PM 8:37 pm 16/05/2026



Academic\_Task\_Tracker | Snyk - Google Chrome  
app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

**snyk**

TENANT  
iramazam2025

GROUP  
iramazam2025

ORGANIZATION  
iramazam2025

Dashboard  
Projects  
Integrations  
Members  
Settings

Product updates  
Help  
Iram Azam

iramazam2025 > Projects > Academic\_Task\_Tracker

Academic\_Task\_Tracker

Overview History Settings

CVSS 4 EXPLOIT MATURITY

- Attacked 0
- Proof of concept 0
- Not defined 0

STATUS

- Open 4
- Patched 0
- Ignored 0

Introduced through requests@2.32.3  
Fixed in requests@2.32.4  
Show more detail

Exploit maturity PROOF OF CONCEPT

Ignore

**M requests** - Insecure Temporary File [🔗](#)

VULNERABILITY | ...

SCORE 419

Introduced through requests@2.32.3  
Fixed in requests@2.33.0  
Show more detail

Exploit maturity NO KNOWN EXPLOIT

Learn about this type of vulnerability

Ignore

30°C Clear 10:47 pm 16/05/2026

Academic\_Task\_Tracker | Snyk - Google Chrome  
app.snyk.io/org/iramazam2025/project/33ee9e9c-8939-4781-9241-bef066905912

**snyk**

TENANT  
iramazam2025

GROUP  
iramazam2025

ORGANIZATION  
iramazam2025

Dashboard  
Projects  
Integrations  
Members  
Settings

Product updates  
Help  
Iram Azam

iramazam2025 > Projects > Academic\_Task\_Tracker

Academic\_Task\_Tracker

Overview History Settings

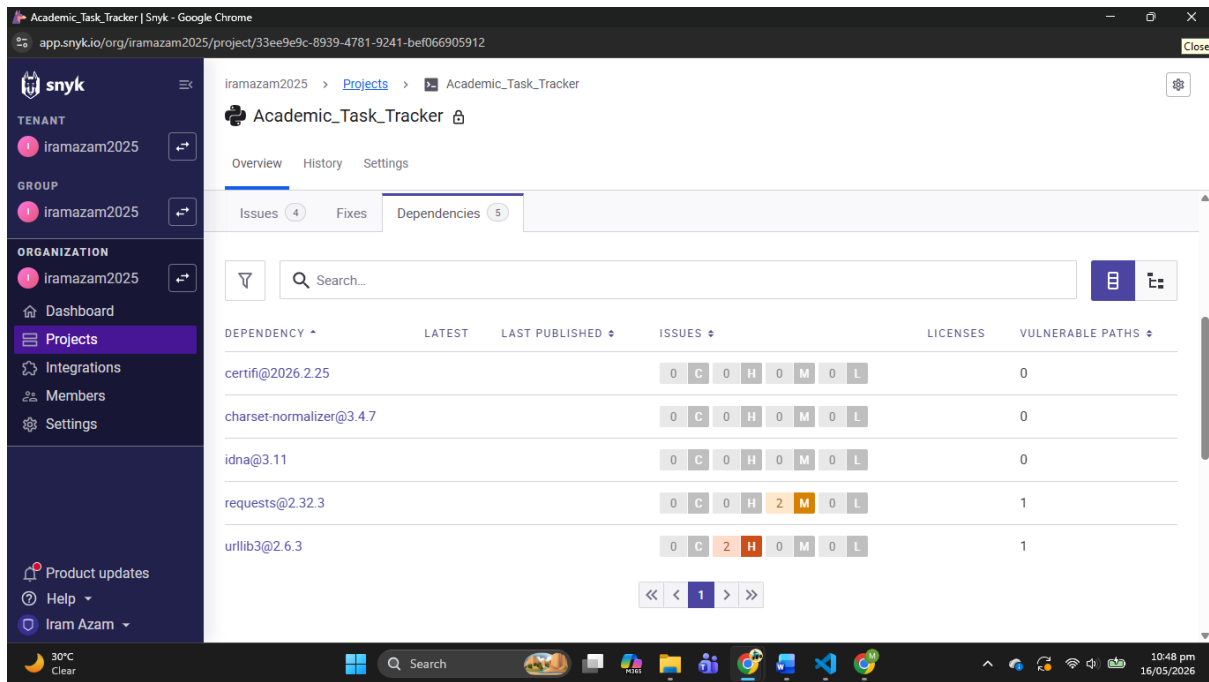
Issues (4) Fixes Dependencies (5)

Fix Advice

Upgradable issues (2)

- > Pin transitive dependency urllib3@2.6.3 to version urllib3@2.7.0
- > Upgrade direct dependency requests@2.32.3 to version requests@2.33.0

30°C Clear 10:47 pm 16/05/2026



## Security Issues Found in Snyk

### Issue 1: urllib3 – Decompression Bomb

This vulnerability may cause memory issues.

#### Fix:

Upgrade urllib3 to latest secure version.

### Issue 2: urllib3 – Sensitive Information Exposure

Sensitive information may be exposed.

#### Fix:

Use updated secure package version.

### Issue 3: requests – Sensitive Information Exposure

Data may become insecure during transmission.

#### Fix:

Upgrade requests package.

### Issue 4: requests – Insecure Temporary File

Temporary files may be insecure.

#### Fix:

Use latest secure package version.

# GitHub repository screenshots:

